Chapter 2 exercises

DS02-E01 • CSR and transpose

A 3×4 matrix has entries `(0,1,5),(1,0,-2),(1,3,7),(2,1,4)`. Give all three CSR arrays. Use counts and prefix sums to construct the transposed COO, listing every output position. Transpose again to check the result.

Hint. Keep immutable starting positions separate from mutable write cursors.

DS02-E02 • Four-dimensional addresses and capacity

A row-major array has dimensions $D_0 \times D_1 \times D_2 \times D_3$, zero-based indices, and w bytes per element. Derive the element and byte offsets of (i_0, i_1, i_2, i_3). State the bounds to check before allocation.

Hint. Interpret successive dimensions as mixed-radix digits.

DS02-E03 • KMP trace

Compute the prefix function of `P=aaaaab`. Find its first match in `T=aaaaaaaaab`, listing q updates after mismatches. Compare the repeated work of naive matching on longer texts of the same form.

Hint. Identify reusable prefixes and suffixes inside the matched segment.

DS02-E04 • Does CSR always save space?

Use 4-byte indices and 8-byte values. Ignore container headers and matrix dimensions. For R rows and k nonzeros, derive COO and CSR storage in bytes and the condition under which CSR is smaller. Interpret `R=1,000,000,k=100`.

DS02-E05 • Row-major addresses and traversal

A contiguous 3×4 matrix is row-major, each element occupies 8 bytes, and the first element starts at b. Find the element offset and address of (2,1). Compare strides when the inner loop traverses columns versus rows; explain why the strides alone do not determine a measured time ratio.

DS02-E06 • Empty rows and CSR invariants

A 4×3 matrix has nonzero entries (1,2,5) and (3,0,−1). Give the three CSR arrays and every row's half-open interval. If row_ptr=[0,0,1,1,3] but there are only two nonzeros, identify the violated invariant.

DS02-E07 • Normalize before interpreting

The 3×3 COO input contains (2,0,4), (0,1,5), (0,1,−5), (2,0,−1). Sum equal coordinates, discard zeros, and give the canonical row-major representation and its transpose. Why does treating all four as distinct nonzero entries violate this chapter's contract?

DS02-E08 • vector size and invalidation

Suppose `vector<int> v={4,7}` and `v.capacity()==2`. Save `p=&v[0]`, then call `v.reserve(10)`. Give the resulting size, determine whether p may still be dereferenced, and state whether `v[2]` has become a valid element.

DS02-E09 • Bytes and code points

The string `A你B` is valid UTF-8 with no additional characters. Find its byte length and the 0-based byte and code-point offsets of `B`. Why is a byte offset not generally a user-perceived character index?

DS02-E10 • Three-dimensional row and column order

A $2 \times 3 \times 4$ array uses 0-based indices and four-byte elements. For coordinate $(1, 2, 1)$, compute element and byte offsets under row-major and column-major layouts. Identify the fastest-changing index in each layout.

DS02-E11 • Overlapping KMP matches

Find every occurrence of pattern `aba` in text `ababa`. Give the prefix function and all starting positions. Why should a search for all occurrences set $q = \pi[M - 1]$ after the first match rather than clear it to 0? What would clearing it miss here?

DS02-E12 • Empty dimensions and transpose output

A COO matrix has dimensions 0×4 and no nonzero entries. Give the transposed dimensions and entries, and the row_ptr if the output is stored in CSR. Why do no nonzeros not eliminate the need to preserve dimensions and row pointers?

References and attribution

Lin Cing, handwritten data structures notes; C++ draft vector.overview and vector.capacity; Knuth, Morris, Pratt, Fast Pattern Matching in Strings (1977).

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https://github.com/xinhi6716/computer-science-self-study

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